

# DEVOPS LAB

## TW1.1 — Git Workflow & Collaboration

|                              |                       |
|------------------------------|-----------------------|
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| <b>Submission Branch</b>     | abhirami-devops       |
| <b>Submission Repository</b> | Devops-Lab-L1_2023-27 |
| <b>Project Repository</b>    | hello-world-flask     |
| <b>Git Feature Branch</b>    | feature/user-auth     |

This document records the step-by-step execution of Assignment TW1.1 and provides screenshot evidence for the Git workflow, branching, conflict creation, conflict resolution, and final GitHub verification.

# 1. Assignment Overview

Objective: demonstrate a complete Git workflow for a simple Hello World Python Flask application, including repository initialization, committing to main, feature branching, a deliberate merge conflict, manual conflict resolution, and pushing the final main branch to GitHub.

## Required workflow

**Task 1.1:** Initialize the Flask project repository, add the initial code, commit it to main, and push it to GitHub.

**Task 1.2:** Create feature/user-auth, modify the application, commit the change, and push the branch.

**Task 1.3:** Modify the same line differently on main, merge the feature branch to generate a conflict, resolve it manually, commit the resolution, and push main.

## 2. Create the Flask Application

A simple Flask Hello World application was created with app.py and requirements.txt. The application is configured to listen on port 5000.

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS>mkdir hello-world-flask

C:\Users\vijayan\OneDrive\Desktop\DEVOPS>cd hello-world-flask

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>dir
Volume in drive C is Windows
Volume Serial Number is 9C88-68EB

Directory of C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask

12-08-2026  15:13    <DIR>          .
12-08-2026  15:14    <DIR>          ..
12-08-2026  15:13                178 app.py
12-08-2026  15:14                 5 requirements.txt
                2 File(s)              183 bytes
                2 Dir(s) 27,492,945,920 bytes free

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git init
Initialized empty Git repository in C:/Users/vijayan/OneDrive/Desktop/DEVOPS/hello-world-flask/.git/

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git branch -M main
```

Flask project files created

### 3. Initialize the Local Git Repository

The local hello-world-flask directory was initialized as a Git repository and the branch was renamed to main. git status confirmed the initial untracked files.

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git branch -M main

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git add app.py requirements.tx
t

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main

No commits yet
```

Git repository initialized

## 4. Stage and Create the Initial Commit

app.py and requirements.txt were staged and committed with the message "Initial commit - Hello World Flask application".

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.py
        new file:   requirements.txt

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git commit -m "Initial commit
- Hello World Flask application"
[main (root-commit) 353725f] Initial commit - Hello World Flask application
 2 files changed, 11 insertions(+)
 create mode 100644 app.py
 create mode 100644 requirements.txt

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main
nothing to commit, working tree clean
```

Initial files staged and committed

## 5. Connect the Local Repository to GitHub

The local repository was connected to abhiraminair2004/hello-world-flask and main was pushed to GitHub.

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git remote add origin https://github.com/abhiraminair2004/hello-world-flask.git

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git remote -v
origin  https://github.com/abhiraminair2004/hello-world-flask.git (fetch)
origin  https://github.com/abhiraminair2004/hello-world-flask.git (push)

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 423 bytes | 38.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/abhiraminair2004/hello-world-flask.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Local repository connected and pushed

## 6. Create feature/user-auth and Commit the Change

A feature branch named feature/user-auth was created. The Flask return statement was modified, staged, committed, and pushed.

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git add app.py

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch feature/user-auth
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

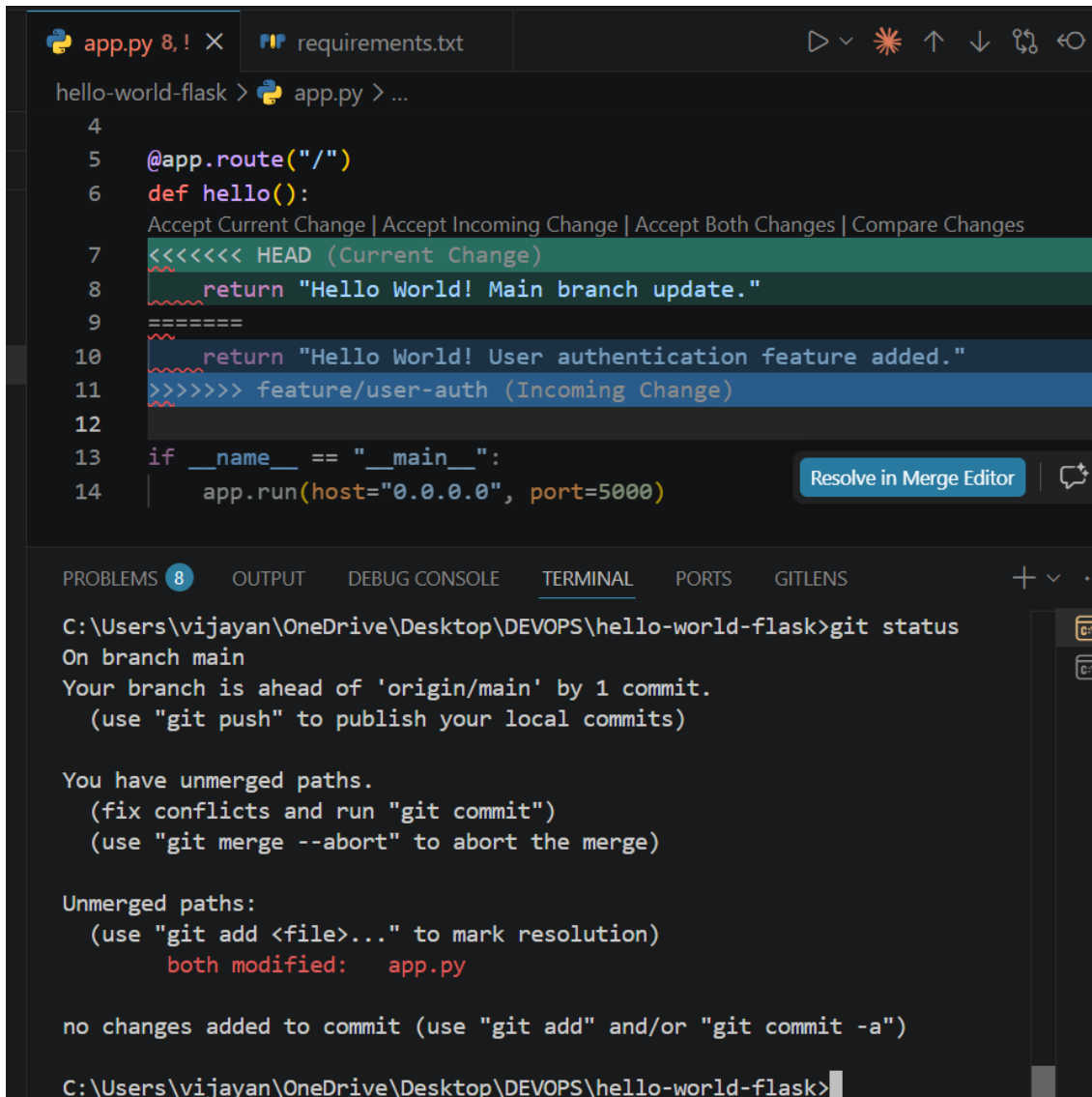
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git commit -m "
Add user authentication feature"
[feature/user-auth ac3d54a] Add user authentication feature
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>
```

Feature branch commit

## 7. Create the Deliberate Merge Conflict

The same return statement was changed differently on main. Merging feature/user-auth into main produced a content conflict in app.py.



The screenshot shows a Visual Studio Code editor with a file named `app.py` open. The code is a Flask application with a single route `/` that calls `hello()`. A merge conflict is present in the `hello()` function. The conflict is between the 'HEAD (Current Change)' and 'feature/user-auth (Incoming Change)'. The 'HEAD' change is `return "Hello World! Main branch update."` and the 'Incoming Change' is `return "Hello World! User authentication feature added."`. The editor shows the conflict with wavy lines and a 'Resolve in Merge Editor' button. Below the editor, the terminal shows the output of `git status`, indicating that the branch is ahead of 'origin/main' by 1 commit and that there are unmerged paths for `app.py`.

```
app.py 8, ! X requirements.txt
hello-world-flask > app.py > ...
4
5 @app.route("/")
6 def hello():
7     <<<<<<< HEAD (Current Change)
8         return "Hello World! Main branch update."
9     =====
10    return "Hello World! User authentication feature added."
11    >>>>>> feature/user-auth (Incoming Change)
12
13 if __name__ == "__main__":
14     app.run(host="0.0.0.0", port=5000)
Resolve in Merge Editor

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
    both modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")

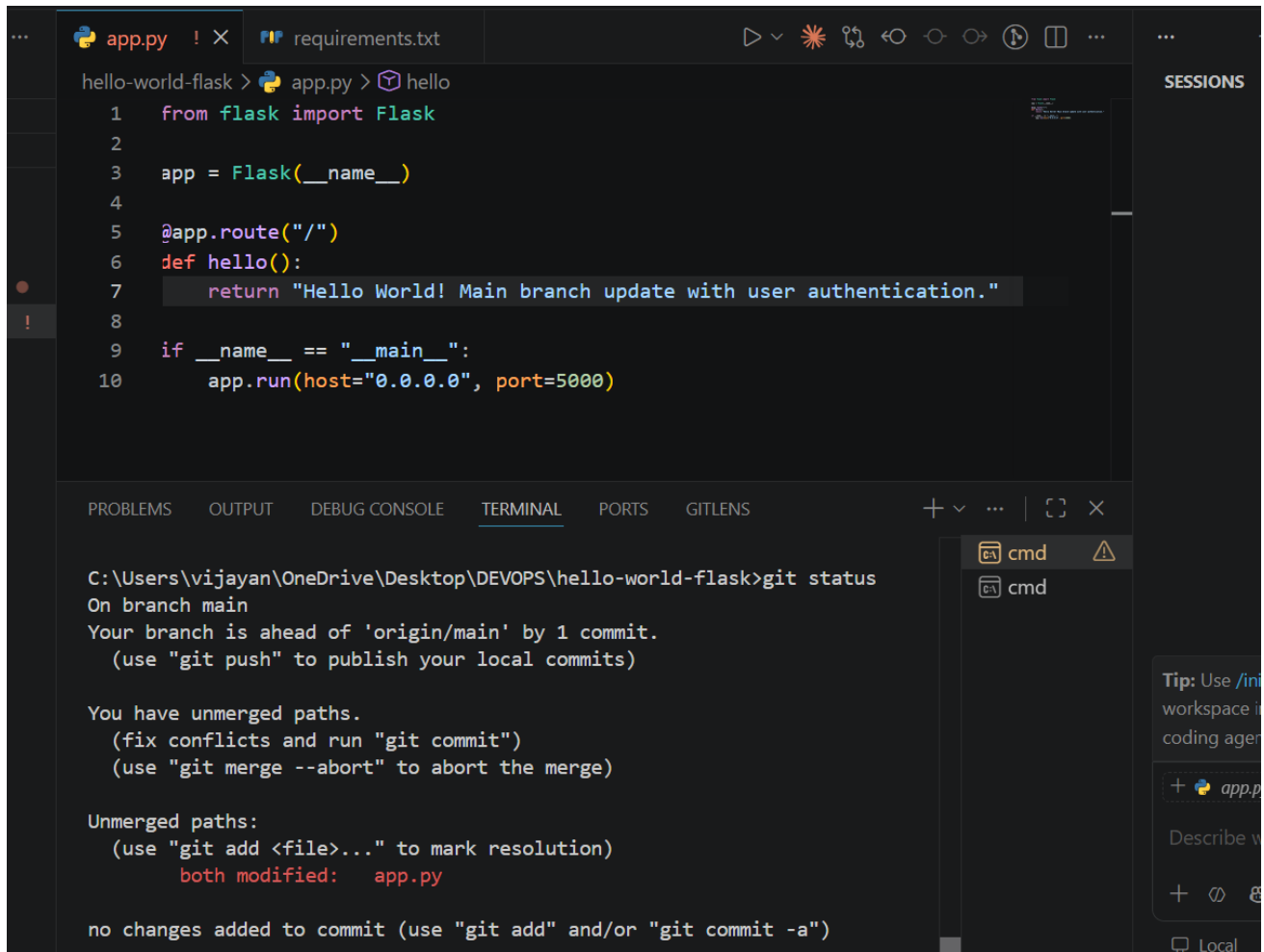
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>
```

Merge conflict generated



## 8. Manually Resolve the Conflict

The conflict markers were removed manually and a combined final message was selected. The resolved file was staged.



The screenshot shows the Visual Studio Code interface. The top editor pane displays the file `app.py` with the following Python code:

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello():
7     return "Hello World! Main branch update with user authentication."
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

The bottom editor pane shows the output of the `git status` command:

```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
```

On the right side of the interface, there is a sidebar with a 'SESSIONS' panel and a 'Local' section. A tip box on the right suggests using `/init` for workspace initialization.

Conflict manually resolved

## 9. Push the Resolved main Branch

The merge commit was completed and the resolved main branch was pushed successfully to GitHub.

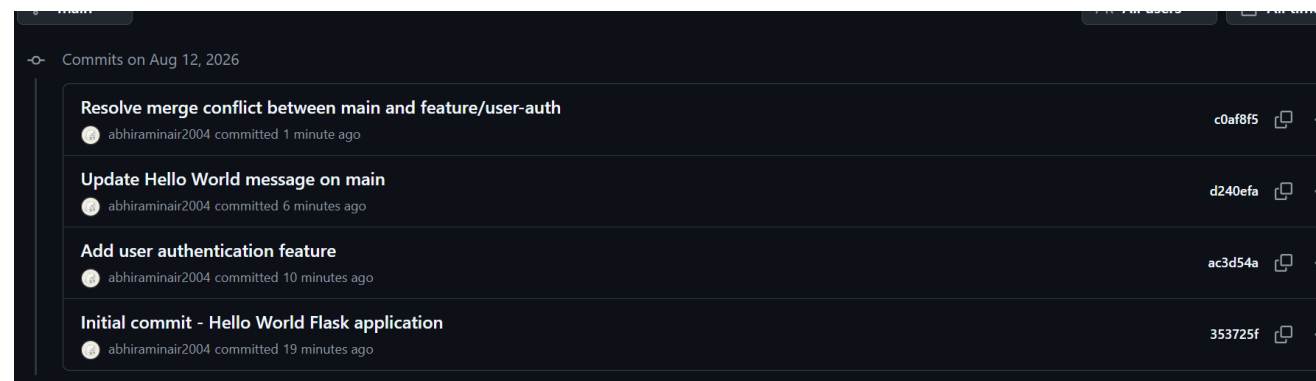
```
C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>git push origin
main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 702 bytes | 234.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/abhiraaminair2004/hello-world-flask.git
353725f..c0af8f5  main -> main

C:\Users\vijayan\OneDrive\Desktop\DEVOPS\hello-world-flask>
```

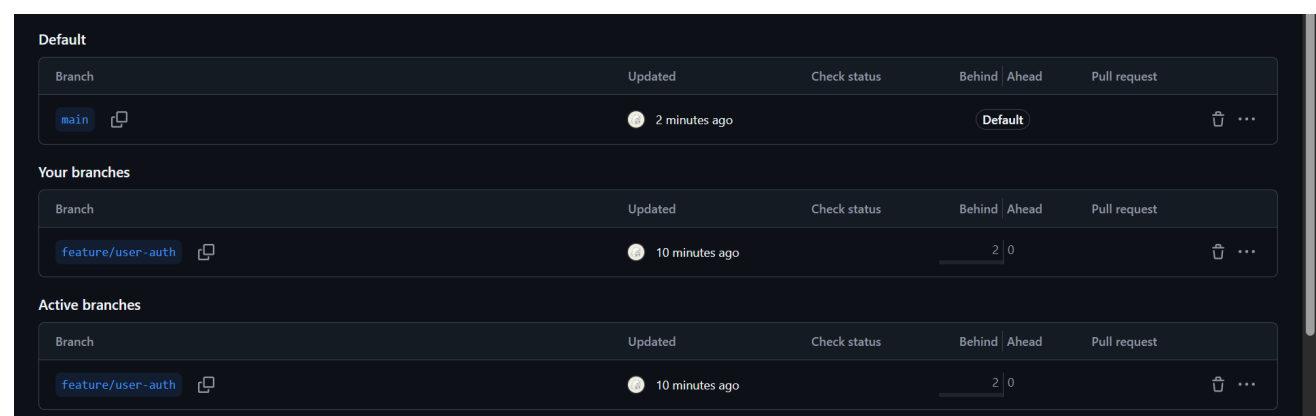
Final main branch pushed

# 10. Verify GitHub Commit History and Branches

GitHub was checked to verify the commit sequence and the presence of both main and feature/user-auth.



Git commit history verified



Branches verified on GitHub

## 11. Final Verification

| Requirement                                                                          | Status    |
|--------------------------------------------------------------------------------------|-----------|
| Task 1.1 — Flask repository initialized and initial code committed to main           | Completed |
| Task 1.2 — feature/user-auth created, modified, committed, and pushed                | Completed |
| Task 1.3 — Same-line conflict created, manually resolved, committed, and main pushed | Completed |
| GitHub main branch verified                                                          | Completed |
| feature/user-auth branch verified                                                    | Completed |
| GitHub commit history verified                                                       | Completed |